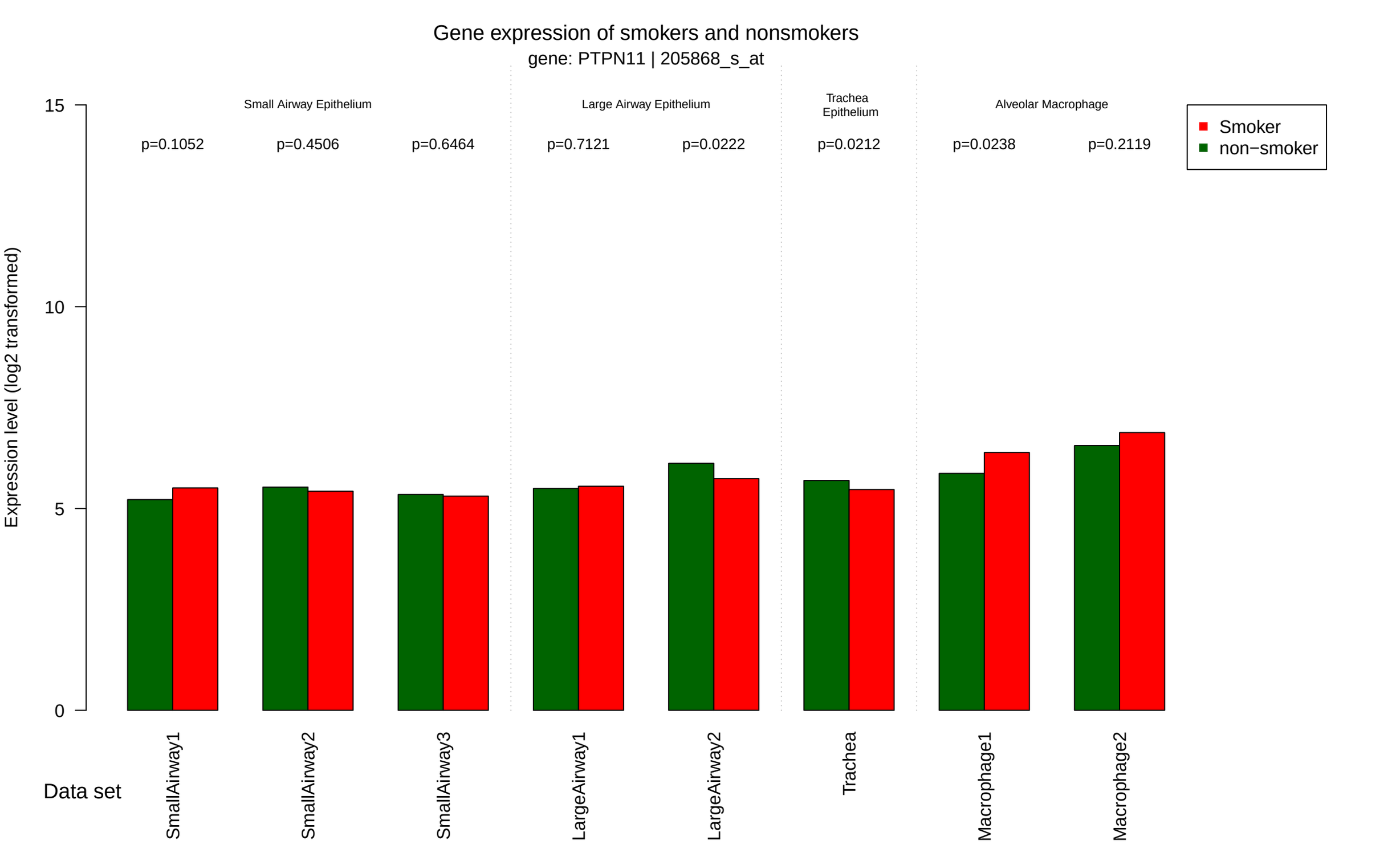
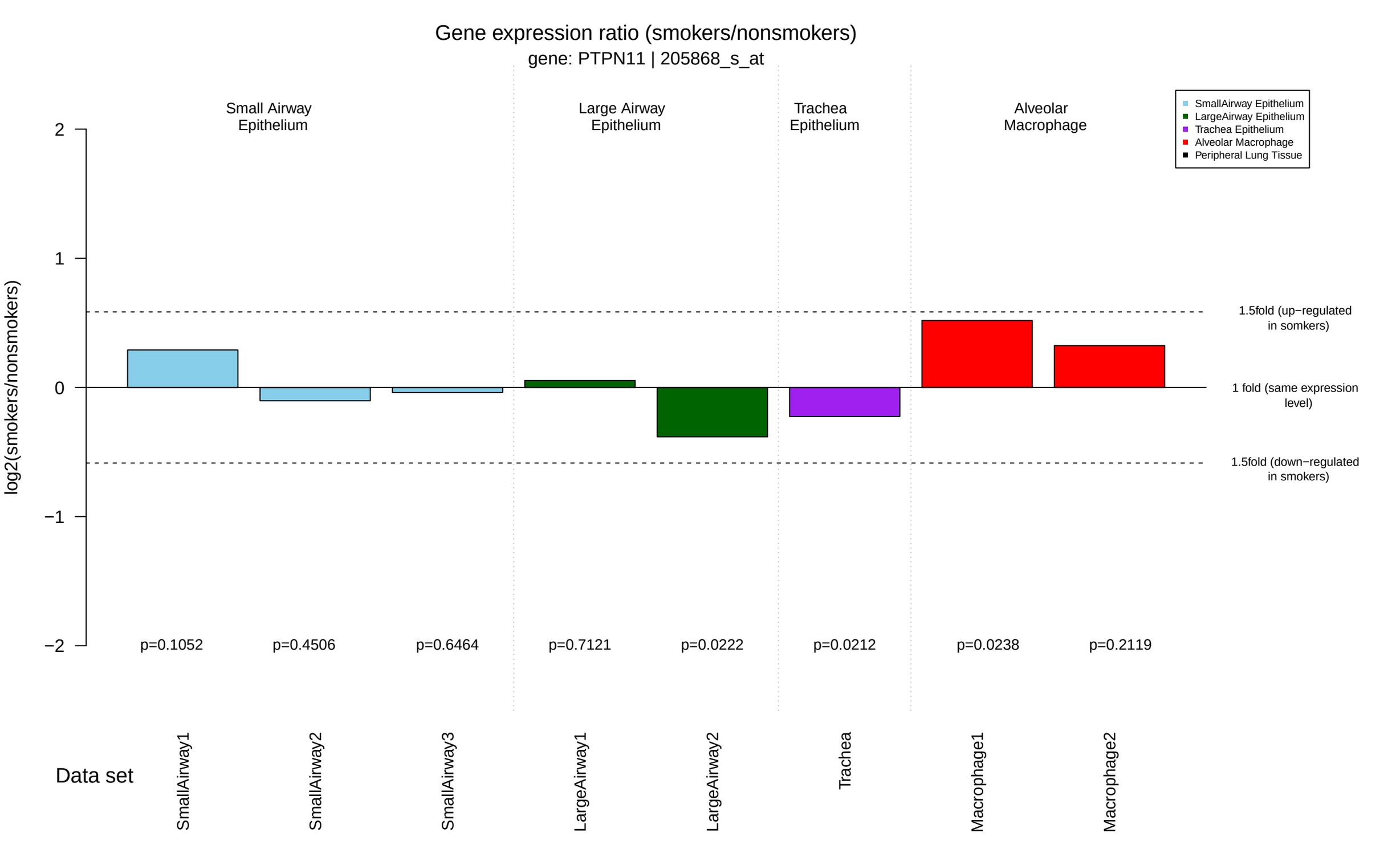
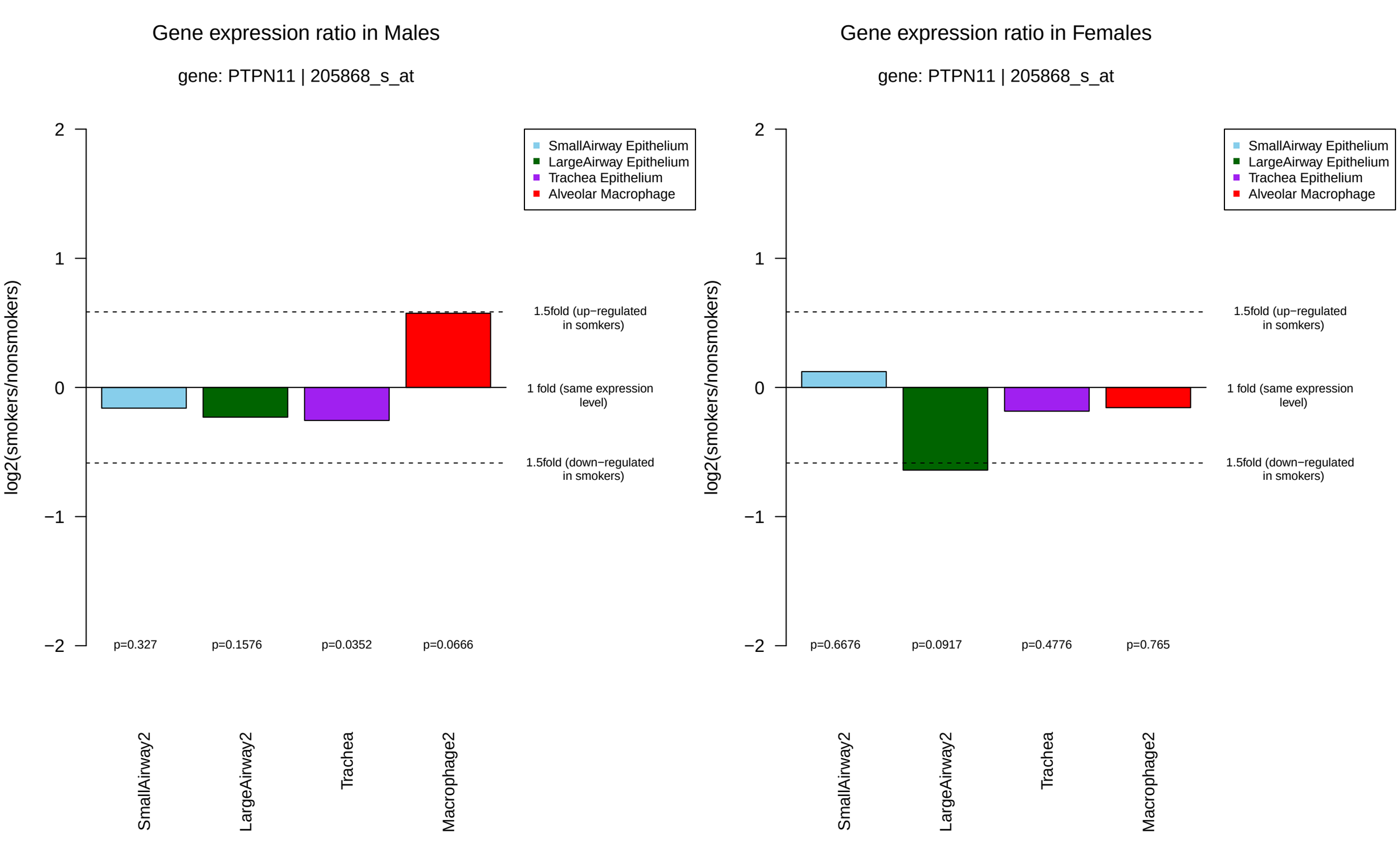
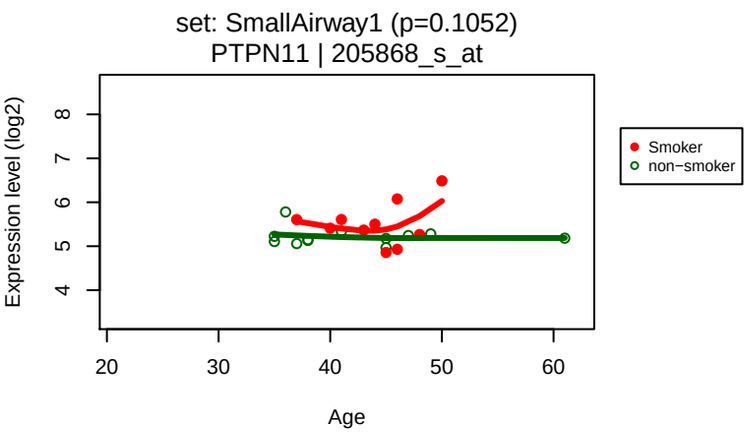
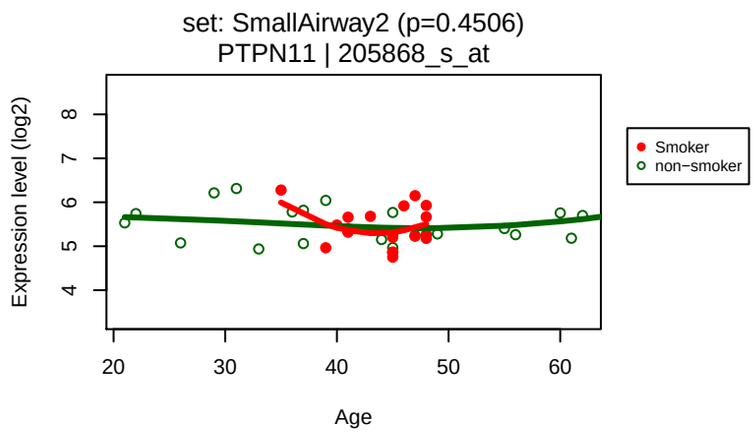
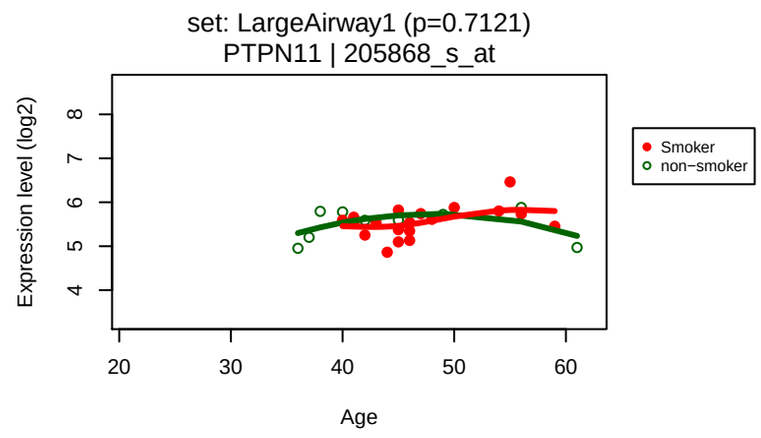

Gene expression of smokers and nonsmokers
gene: PTPN11 | 205868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1052
p=0.4506
p=0.6464
p=0.7121
p=0.0222
p=0.0212
p=0.0238
p=0.2119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

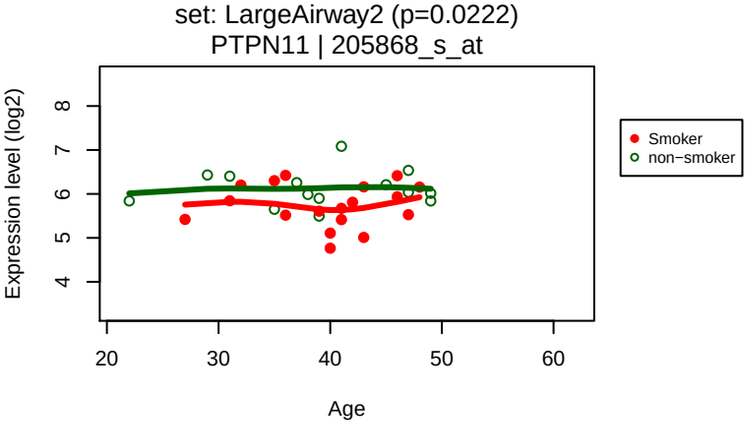

Gene expression ratio (smokers/nonsmokers)
gene: PTPN11 | 205868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1052
p=0.4506
p=0.6464
p=0.7121
p=0.0222
p=0.0212
p=0.0238
p=0.2119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

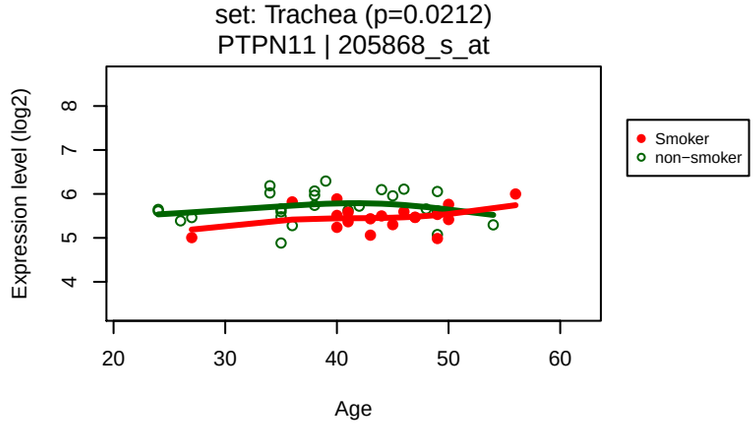

Gene expression ratio in Males
gene: PTPN11 | 205868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.327
p=0.1576
p=0.0352
p=0.0666
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN11 | 205868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6676
p=0.0917
p=0.4776
p=0.765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

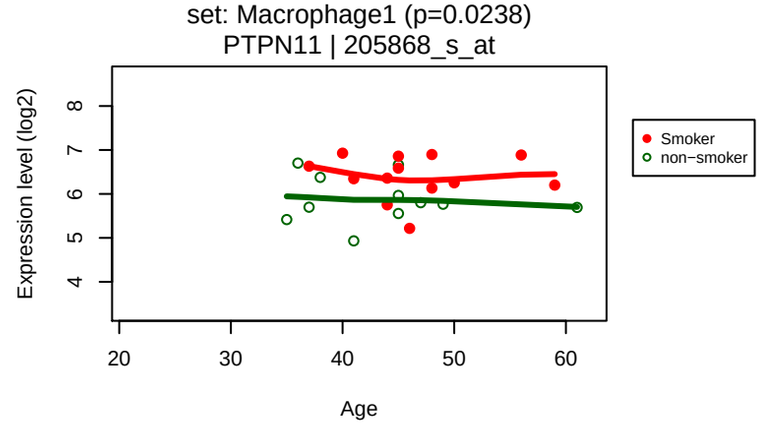

set: SmallAirway1 (p=0.1052)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker

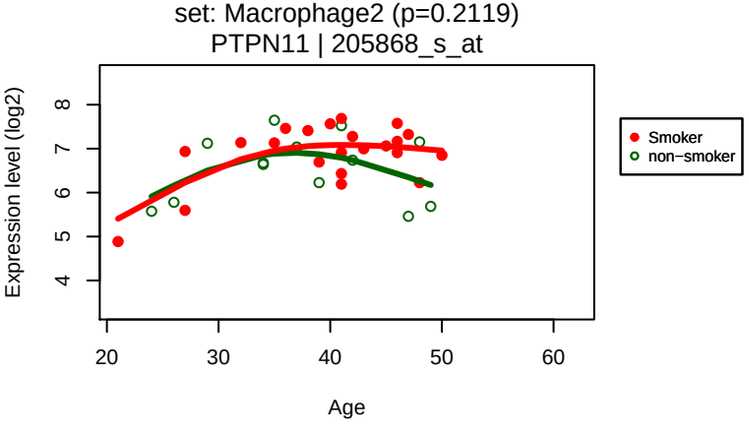

set: SmallAirway2 (p=0.4506)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker

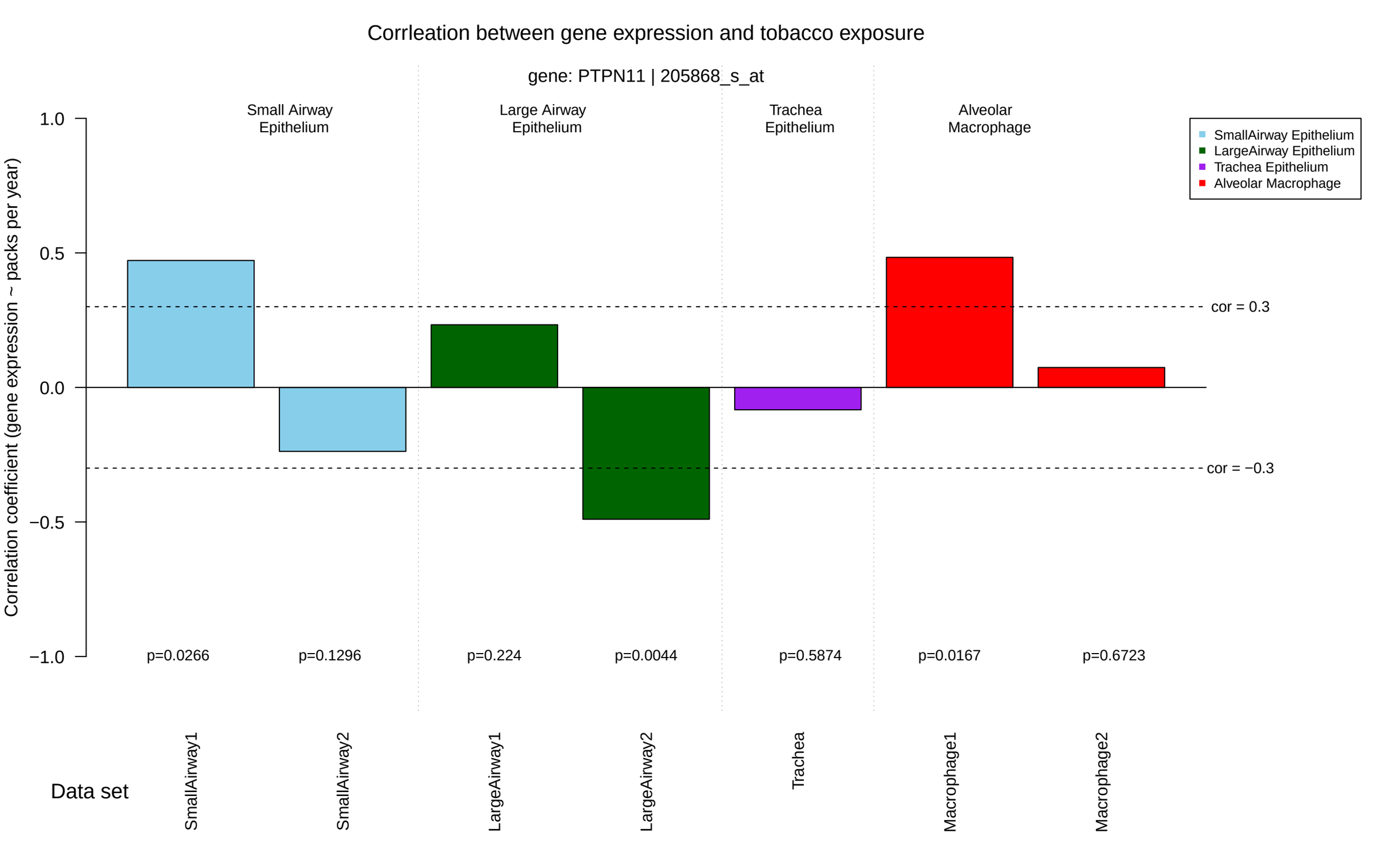

set: LargeAirway1 (p=0.7121)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0222)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0212)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0238)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2119)
PTPN11 | 205868_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN11 | 205868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0266
p=0.1296
p=0.224
p=0.0044
p=0.5874
p=0.0167
p=0.6723
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2